

FABIAN D. HERRERA

fabianherrera73@gmail.com | +1 (438) 630-5373 | Github: [FabianHerr](#)

EDUCATION

McGill University

B.Sc. in Software Engineering - Minor in Entrepreneurship

845 Sherbrooke St W, Montreal, QC

Expected Graduation: 2028

- **Relevant coursework:** Data Structures and Algorithms, Software Systems, Algorithms and Complexity, Computer Systems, Probability and Statistics.

PROFESSIONAL EXPERIENCE

Transport Laberge

Data Engineering & Analytics Intern

Montreal, QC

January 2026 – June 2026

- Built scalable **ETL pipelines** processing **20M+ records** using **Python** and **SQL**, removing manual ingestion steps and enabling reliable, production-ready ML data flows.
- Working on an **AI voice agent** that connects **OpenAI APIs** with internal transport systems to answer operational questions, extract logistics information, and trigger workflows, with the goal of reducing dispatcher manual workload by **at least 25%** by the end of the internship.
- Operated in an Agile workflow with daily stand-ups, delivering targeted features each sprint and tracking progress in Azure DevOps.
- Built **Power BI** reports visualizing key KPIs (FTFR, MTBF, TCO) to drive data-based decisions.
- Deployed an API on **Vercel** (Vapi, Supabase) to automate post-call email workflows.

PROJECTS & RECOGNITIONS

Logbait™ – Cloud-Integrated Roguelike Game

Team Project (8) | McGill

Montreal, QC

January 2026 – Now

- Engineered core gameplay systems including **player movement, combat mechanics, and enemy AI behaviors**, implementing targeting, attack logic, and health/damage systems in **C#**.
- Designed a **modular ability system** enabling reusable gameplay powers with cooldown logic, visual effects, and state management.
- Built **responsive UI elements** (health bars, ability indicators, combat feedback) to improve real-time player interaction.
- Collaborated in an **Agile Git-based workflow** using feature branches and merge conflict resolution to integrate gameplay systems into a shared codebase.

Swap - P2P Currency Exchange Web App | [GitHub](#)

Personal Project

Montreal, QC

July 2025 – Now

- Built a **full-stack web app** enabling local cash currency exchanges without intermediaries.
- Developed **Node.js/Express** backend with **MongoDB/Mongoose** for user accounts, offers, and transactions.
- Responsive **React UI** (Bootstrap) with **JWT auth**.

House Price Prediction Model | [Certificate](#) | [Car Model](#)

Dawson AI Research and Development Project

3040 Sherbrooke St W, Montreal, QC

August 2023 - July 2024

- Developed a linear regression model to predict used car prices with **~80% accuracy (R-squared)**.
- Cleaned **50,000+ entries**, cutting error rates by **20%**, handling missing values and outliers.
- Evaluated the model using **MSE** and **MAE**, achieving a low MSE indicating accurate predictions.
- Built with **Python** using **Pandas**.
- Collaborated with my team to develop a house price prediction model using linear regression techniques.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C#, HTML, CSS, R, Bash/sh

Frameworks & Libraries: React, Node.js, Express, Mongoose, Bootstrap, Pandas, NumPy, scikit-learn

Database: MongoDB

Tools & Technologies: Git/Github, Jupyter Notebook, Unity, Visual Studio Code, Canva, Figma, Azure DevOps, Power BI, Power Automate, SharePoint, Vercel, Supabase, MS Office 365

Concepts: Object-Oriented Programming (OOP), REST APIs, Machine Learning (Linear Regression), Data Preprocessing, Model Evaluation (MSE, MAE), Bash/sh scripting and Linux, Agile/Scrum, CI/CD, DevOps, ETL, KPI Reporting.

Languages: French (Fluent), English (Fluent), Spanish (Fluent)

Others: Possess a valid driver's license and own a car.